

Model Comparison — Holdout 2022–2025 [RMSE loss variant]
RMSE, MAE, Skill vs RW, R², Hit Rate | DM = Harvey et al. (1997) vs RW | DM OLS = Diebold-Mariano vs OLS

Horizon	CatBoost RMSE	CatBoost MAE	CatBoost Skill%	CatBoost R²	CatBoost Hit	CatBoost p(DM)	CatBoost p(DM OLS)	HTBoost RMSE	HTBoost MAE	HTBoost Skill%	HTBoost R²	HTBoost Hit	HTBoost p(DM)	HTBoost p(DM OLS)	Lasso RMSE	Lasso MAE	Lasso Skill%	Lasso R²	Lasso Hit	Lasso p(DM)	Lasso p(DM OLS)	SARIMA RMSE	SARIMA MAE	SARIMA Skill%	SARIMA R²	SARIMA Hit	SARIMA p(DM)	SARIMA p(DM OLS)	OLS RMSE	OLS MAE	OLS Skill%	OLS R²	OLS Hit	OLS p(DM)	OLS p(DM OLS)
0w	0.0650	0.0503	+9.0%	0.171	61.3%	<.001	—	0.0600	0.0472	+16.0%	0.295	66.7%	<.001	—	0.0632	0.0490	+11.5%	0.217	65.8%	<.001	—	0.0673	0.0507	+5.7%	0.110	61.7%	<.001	—	—	—	—	—	—	—	
1w	0.1008	0.0775	+11.6%	0.219	66.7%	<.001	—	0.0994	0.0768	+12.9%	0.241	64.4%	<.001	—	0.0996	0.0761	+12.7%	0.238	65.8%	<.001	—	0.1105	0.0845	+3.2%	0.063	57.2%	0.033	—	—	—	—	—	—	—	
2w	0.1205	0.0940	+12.4%	0.233	66.5%	<.001	—	0.1160	0.0901	+15.7%	0.289	66.5%	<.001	—	0.1183	0.0916	+14.0%	0.259	67.4%	<.001	—	0.1321	0.1031	+3.9%	0.077	66.1%	0.005	—	—	—	—	—	—	—	
1m	0.1394	0.1080	+17.3%	0.315	70.3%	<.001	0.458	0.1411	0.1051	+16.3%	0.299	72.1%	<.001	0.624	0.1415	0.1054	+16.1%	0.295	71.2%	<.001	0.734	0.1611	0.1247	+4.5%	0.087	61.2%	<.001	—	0.1398	0.1059	+17.1%	0.312	67.1%	<.001	—
3m	0.2567	0.2090	+15.1%	0.279	72.4%	<.001	0.904	0.2347	0.1888	+22.4%	0.397	75.7%	<.001	0.724	0.2299	0.1865	+24.0%	0.422	75.2%	<.001	0.795	0.2743	0.2332	+9.3%	0.176	77.6%	<.001	—	0.2179	0.1726	+27.9%	0.480	75.2%	0.005	—
6m	0.2892	0.2426	+20.2%	0.362	74.6%	0.008	0.912	0.2395	0.1980	+33.9%	0.563	87.3%	<.001	0.711	0.2334	0.1890	+35.6%	0.584	85.8%	<.001	0.644	0.3180	0.2755	+12.2%	0.229	82.7%	<.001	—	0.2234	0.1609	+38.3%	0.619	87.8%	0.009	—
12m	0.1910	0.1511	+10.4%	0.192	70.2%	0.090	0.012	0.2054	0.1592	+3.7%	0.066	64.3%	0.308	0.009	0.2204	0.1668	-3.3%	-0.075	51.5%	0.657	0.041	0.2080	0.1633	+2.5%	0.042	53.2%	0.293	—	0.2594	0.2130	-21.6%	-0.490	46.8%	0.988	—